Bipin Lekhak

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Summary

PhD candidate building **scientific machine learning models of dynamical systems**: latent neural-ODE models that combine structural assumptions with data to predict how a system will respond to control inputs and interventions. The models separate a system's intrinsic dynamics from its response to inputs, are identifiable, and adapt to each individual system from its own history through meta-learning. Validated on physical systems observed through video, mechanistic simulators and clinical time series. Before the PhD, four years as a machine learning engineer shipping computer-vision and forecasting systems.

Research interests: scientific machine learning, time-series modelling and forecasting under interventions, latent dynamical systems and neural ODEs, causal inference over time, identifiable representation learning, meta-learning for adaptation across systems, machine learning for health.

Education

Rochester Institute of Technology, Rochester, NY — *Aug 2022 – Present*
Ph.D. in Computing and Information Sciences — *Expected 2027*

- Advisor: Prof. Linwei Wang, Computational Biomedicine Laboratory
- Dissertation proposal (defended July 2026): *Learning Dynamics for Forecasting Future Trajectories under External Interventions*

Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal — *Nov 2014 – Sep 2018*
B.E. in Electronics and Communication Engineering

Publications

Manuscripts under review

iASIDE: Identifiable, Adaptable, and Separable Interventional Dynamics — *2026*
Bipin Lekhak, Yubo Ye, Suraj Poudel, Ruby Shrestha, Linwei Wang

Peer-reviewed conference papers

ProReGen: Progressive Residual Generation under Attribute Correlations — **ICLR 2026**
Ruby Shrestha, Ajay Gopi, Casey Meisenzahl, ***Bipin Lekhak***, Linwei Wang
International Conference on Learning Representations. OpenReview

Novel In-Home Cardiac Monitoring for Heart Failure Patients — **CinC 2023**
Bipin Lekhak, Ryan Missel, Dillon Dzikowicz, Solomiya Rachynska, Wojciech Zareba, Linwei Wang
Computing in Cardiology. IEEE Xplore

Research Experience

Graduate Research Assistant — *Rochester, NY*
Computational Biomedicine Laboratory, RIT (PI: Prof. Linwei Wang) — *Aug 2022 – Present*

- **Interventional latent dynamics (iASIDE), project lead and first author.** Developed a latent neural-ODE framework that predicts a system's trajectory under future control inputs or treatments. It separates intrinsic dynamics from the response to inputs, adapts both to each system from its history through amortized (feed-forward) meta-learning, and establishes identifiability of the learned dynamics. Gains over baselines are largest at long horizons and across heterogeneous systems.
- **Physics-based simulation.** Wrote a simulator of damped, forced pendulums under stochastic inputs, rendered as image sequences, to test whether models recover physical parameters from video. Also evaluated

on a pharmacokinetic–pharmacodynamic simulator of tumour growth under therapy and on MIMIC-III intensive-care records.

- **Reusable research software.** Built the project's experiment framework: 10+ baselines (Causal Transformer, CRN, RMSN, MSM, G-Net, SCIP, TE-CDE, TESAR-CDE, ACTIN) on three benchmarks, with Hydra configuration, PyTorch Lightning, MLflow tracking, SLURM sweeps across GPU clusters, and regression tests that verified a migration from PyTorch 1.13 to 2.x.
- **Real-world sensor data** (NIH R01NR018301, with the University of Rochester Medical Center). Analysed multi-sensor time series from devices deployed in heart-failure patients' homes for 90 days. Characterised deployment failures and missing data, built data dashboards, and predicted 21-day hospitalisation from 7-day windows (AUC 0.74).
- **Treatment-effect estimation over time** (work in progress, with S. Poudel). Temporal X- and R-learners that build on the separable latent dynamics to identify causal effects.

Industry Experience

Machine Learning Engineer *Kathmandu, Nepal*
Fusemachines *Sep 2018 – Jun 2022*

- Built deep-learning semantic segmentation for medical images with U-Net, Mask R-CNN and Detectron.
- Reconstructed 3D anatomical structures from 2D images by gradient-free optimisation over statistical shape models.
- Forecast product sales from historical data with a multi-stage model, each stage encoding a different assumption about the sales pipeline (LSTMs, Prophet, gradient-boosted trees); designed and tested Bayesian variants in PyMC.

Teaching

Teaching Assistant, GCCIS-820 Quantitative Foundations, RIT *Fall 2025*
PhD core course: linear algebra, optimisation and statistical inference

- Supervised and graded team projects and reports; built the tool used to form project teams.

Skills

Scientific ML and time series: neural ODEs/CDEs, latent state-space models, physics-based simulation, meta-learning, variational inference, identifiability, Bayesian modelling, forecasting (LSTMs, Prophet, gradient-boosted trees)

Causal ML: potential outcomes, time-varying confounding, IPW and marginal structural models, meta-learners (X- and R-learners)

ML stack: Python, PyTorch, PyTorch Lightning, torchdiffeq, Hydra, MLflow, Ray Tune, scikit-learn, PyMC, pandas, NumPy; computer vision (U-Net, Mask R-CNN, Detectron)

Infrastructure: SLURM and HPC clusters, Linux, Podman/containers, self-hosted MLflow, uv, pytest, Git

Other: R, MATLAB, SQL, LaTeX, Streamlit, Dash